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Report (Sales Data Analysis)

Data Science Report

Sales Data

Analysis

Dataset Discussion

Data Source and Description: The dataset for my analysis was sourced from an online open data platform (Kaggle). It can be referenced via: "https://www.kaggle.com/datasets/gajendrasharmagl/sales-data." The dataset contains approximately **10,000** records of individual sales transactions (more than the minimum 500 records required). Each record represents a single order with multiple variables captured. There are at least **8** columns (variables) in the dataset, including:

- **OrderID**: a unique identifier for each transaction.
- **Product**: the name of the product sold.
- **Category**: the category or department of the product (e.g., Electronics, Books, Apparel).
- **Region**: the geographical region of the sale (e.g., North America, Europe, Asia).
- **Quantity**: the number of units sold in that transaction (numeric).
- **PricePerUnit**: the price for a single unit of the product (numeric).
- **TotalSales**: the total revenue from the transaction (calculated as Quantity × PricePerUnit).
- **Date**: the date of the sale (timestamp, covering late 2024 to early 2025).

These variables include a mix of data types: **qualitative** data (Product, Category, Region), and **quantitative** data (Quantity, PricePerUnit, TotalSales), date/time data (Date). This rich dataset provides a historical snapshot of sales, allowing for analysis as well as comparisons.

Hypothesis and Key Business Problems

Based on my initial analysis of the sales data, I propose the following hypothesis: "By analysing sales data, demand forecasting and inventory control will increase, resulting in greater revenues and lower costs." From this hypothesis, I can point out the following three business questions:

1. What drives daily sales?

- Management needs to know what drives daily sales. If sales vary by product category, by date, or by region, knowing which one drives sales more—product category, for example, seasonality (from trends based on date) or related trends to specific regions—can help management better market and better forecast inventory. If they know they can have more resources allocated at certain times or to certain categories that drive better sales, they can increase revenues. For example, if they find out that they have more sales on weekends than weekdays or that one category has a lot of promise, more resources can focus on those findings. [1]

2. What are the future sales trends of the best-selling products?

- Improved demand predictions of best sellers improve inventory efficiency and decrease lost sales. If the company could predict future sales of its best-selling products with certain accuracy (it has quantity sold, pricing changes, and date trends from the database of its previous sales), the company would know how to always stock these items at the appropriate level. There would be decreased stockouts (no lost sales revenue) and decreased overbuying (decreased carrying costs), impacting the financial bottom line. [1]

3. Which areas are consistently underperforming that stock could be reallocated?

- If areas are consistently underperforming, management can try to fix the problem with specific targeted promotions or reallocation of inventory. For example, if an entire area (some countries or a continent) is consistently operating with low sales in comparison to overall increase (even if some country or area in that region is sustaining the numbers), they can tell if this is lack of company penetration in the market or if the products that exist there are not suitable for buyer needs in that region. In this instance, the company can

avoid wasting time and money on products that are not generating revenues and instead, use that stock for other areas generating positive results or use that information for more personalized stock adjustment for those underperforming. The company does not want to have too much cash tied up in non-moving inventory. [1]

All these issues come from a number-based perspective for micro and macro management adjustments. Solving such issues will improve demand forecasting, improve inventory management policies as well as target those areas with the highest managerial potential for growth. [1]

Data Preprocessing

Data Preprocessing (Excel):
Before any sort of analysis or modelling, I must ensure proper data quality with a preprocessing step in Excel. The following preprocessing was done:

Missing Values:

- Determine whether any fields/columns have missing/null values. For example, see if the number of products ordered is empty/missing, or if sales or OrderID is empty. In my case, I had no major missing values in my columns of concern—I would have had to either impute with other values (i.e., taking the average sales of like products and imputing for blanks where sales are missing) or deleting full rows or transactions to keep us from entering information that was guesswork. When too much information is not available, it is better to remove it from the dataset because it will skew my analysis and fill it with inaccuracies based on empty cells. [2]

Duplicates:

- Determine whether duplicates exist in data. Each OrderID must be unique—or else there are duplicate transactions. If there are duplicate entries (i.e., exact same transactions appearing more than once), I delete them to ensure that sales aren't counted twice in analysis. This way, summary statistics such as total revenue or average daily sales remain accurate. [2]

Correcting Data Types:

- Data types corrected in a number of places - dates became date (to allow time-based functionality like sorting by date or generating a month/day), the numeric fields of Quantity, PricePerUnit, and TotalSales became number (to allow addition/subtraction). This is important for the corrections of type to work via Excel, Python to give proper indication of how to calculate or treat each field. [2]

Treating Outliers and Inconsistencies:

- I cross-referenced the TotalSales fields with each other and other numeric fields to see if there were outliers or miscalculations. For example, in the raw export, there were a few FinalTotal fields that did not equal Quantity * PricePerUnit (i.e. a purchase of 12 units with 1800 each had 7856.04 as a total which seems either an entry error or discount not applied that was not mentioned). These fields were called out - some recast to allow for the correct calculation - some dropped from FinalTotal if it seemed to be an error. In addition, I called out any fields that were outliers constituting extremely high sales transactions or zero sales (which would throw off analysis). [2]

What I did: (Data Cleaning)

- Such as regularized spelling and formatting of categorical fields (i.e. not "NA" and "North America" but all one way or the other); excess space removal; etc.; making sure fields were within expected range of dates (this project only would expect late 2024 through early 2025). [2]

Once this was done, the data was in a trustworthy state for analysis.

(Screenshot 1: Raw dataset sample with initial anomalies)

OrderID	Product	Category	Region	Quantity	PricePerU	TotalSales	Date
ORD10000	Smartwatch	Electronics	North America	1	300	300	24/09/2024
ORD10001	Science	Books	Asia	4	30	120	18/01/2025
ORD10002	Running Shoes	Sports	Australia	1	100	100	15/01/2025
ORD10003	Washing Machine	Home Appliances	North America	2	1800	3600	29/11/2024
ORD10004	Camera	Electronics	Asia	2	700	1400	01/11/2024
ORD10005	Smartphone	Electronics	Asia	1	800	800	14/12/2024
ORD10006	Sofa	Furniture	Asia	2	5000	10000	20/01/2025
ORD10007	Sofa	Furniture	Asia	1	5000	5000	14/01/2025
ORD10008	Washing Machine	Home Appliances	Europe	1	1800	1800	31/10/2024
ORD10009	Vacuum Cleaner	Home Appliances	North America	1	500	500	24/09/2024
ORD10010	Smartphone	Electronics	Asia	2	800	1600	22/02/2025
ORD10011	Board Game	Toys	Asia	1	80	80	11/01/2025
ORD10012	Laptop	Electronics	Europe	1	1200	1200	17/02/2025
ORD10013	RC Car	Toys	Europe	1	150	150	03/10/2024
ORD10014	Jeans	Apparel	Asia	2	50	100	05/11/2024
ORD10015	Washing Machine	Home Appliances	Asia	12	1800	7856.044	08/12/2024
ORD10016	Philosophy	Books	North America	14	18	56.67804	01/10/2024

Screenshot 2: Cleaned dataset after preprocessing).

	OrderID	Product	Category	Region	Quantity	PricePerU	TotalSales	Date
0	ORD10000	Smartwatch	Electronics	North America	1	300	300	24/09/2024
1	ORD10001	Science	Books	Asia	4	30	120	18/01/2025
2	ORD10002	Running Shoes	Sports	Australia	1	100	100	15/01/2025
3	ORD10003	Washing Machine	Home Appliances	North America	2	1800	3600	29/11/2024
4	ORD10004	Camera	Electronics	Asia	2	700	1400	01/11/2024
5	ORD10005	Smartphone	Electronics	Asia	1	800	800	14/12/2024
6	ORD10006	Sofa	Furniture	Asia	2	5000	10000	20/01/2025
7	ORD10007	Sofa	Furniture	Asia	1	5000	5000	14/01/2025
8	ORD10008	Washing Machine	Home Appliances	Europe	1	1800	1800	31/10/2024
9	ORD10009	Vacuum Cleaner	Home Appliances	North America	1	500	500	24/09/2024
10	ORD10010	Smartphone	Electronics	Asia	2	800	1600	22/02/2025
11	ORD10011	Board Game	Toys	Asia	1	80	80	11/01/2025
12	ORD10012	Laptop	Electronics	Europe	1	1200	1200	17/02/2025
13	ORD10013	RC Car	Toys	Europe	1	150	150	03/10/2024
14	ORD10014	Jeans	Apparel	Asia	2	50	100	05/11/2024
15	ORD10015	Washing Machine	Home Appliances	Asia	12	1800	21600	08/12/2024
16	ORD10016	Philosophy	Books	North America	14	18	252	01/10/2024

Discussion of the Technique: Linear Regression

Chosen Technique – Linear Regression:
As the analytical modelling technique, I selected Linear Regression as the core technique. Linear regression is a supervised learning method for predicting a continuous outcome variable from one or more predictor variables. In this scenario, the continuous outcome I would like is typically a measure of sales (say, total daily sales or total sales of a particular product), and the predictor features can be factors like product category, quantity sold, region, and time (date or day-of-week). The goal of linear regression is to find the best-fit linear equation (a straight-line relationship in multidimensional space) describing the relationship of the predictors quantitatively with the target variable. [3]

Why Linear Regression is Appropriate:

The nature of the target variable – sales revenue – is numeric and continuous, and linear regression is thus the appropriate choice. I am asking questions of “how much” (say, how much will the sales increase by, or what will future sales be), which is the kind of question that is answered by regression analysis rather than classification. As a concrete example, for the task of forecasting future sales patterns for top-performing products, a regression model can be trained on historical sales data (say, aggregated by day, by month, etc.) with features like time period, promotion indicators, and product category to project the amount of sales for future periods. The linear regression model will make a continuous prediction (say, next month’s sales in dollars), which is a direct answer to a forecasting question. [3]

Furthermore, linear regression is valuable because it also provides interpretable coefficients for each feature. This includes not only being capable of making predictions for future sales but also being capable of seeing the impact of each variable on sales. For example, the model can display the level of increased sales for each additional unit sold (Quantity coefficient), or the impact of being located in a given region on revenue compared with some other region (if Region is dummy coded as variables on a multiple regression). The interpretability is also linked with my original business problem of identifying key drivers of the sales performance, as the size and significance of the regression coefficients can reflect which variables (like seasonality, specific categories, etc.) contribute the most toward variation of the sales. [3]

Feature Usage in the Model: In linear regression, some of the cleaned dataset variables will serve as input features:

- Region and Product Category:
  These can be expressed in numerical terms (i.e., one-hot encoding for the category or region) for the model to work with. With these, the model is able to make predictions based on the popularity of the category or regional variations of demand. [3]
- Quantity and Price:
  These numerical fields relate with total sales (TotalSales = Quantity * PricePerUnit). TotalSales could be the target, but if I was predicting total daily or total monthly sales, the sum of the quantity or the average of the price could be predictor variables. In order-level revenue forecasting, quantity and price would be the most obvious predictors of the sale of that order. [3]
- Date/Time Features:
  The date can be transformed into features like day of the week, month, or if it was a holiday period, in order to be able to capture seasonal patterns. As a for-instance, the inclusion of a feature for the month may enable the model to learn the November/December spikes and the January lulls. [3]

My linear regression choice is also justified by the data preparation: after preprocessing, the dataset is error-free and free of significant missing values, thus easier for the regression model to function. Linear regression also assumes linear association of the predictors with the target; even if real-life sales can be intricate, as a starting point, it provides a straightforward model that is easy to deploy, rapid to train, and straightforward for stakeholders to interpret. It is a default approach for predicting continuous results like sales, and it serves as the basis for the construction of more complex models if necessary. Generally, linear regression is appropriate for this project because it directly addresses the predictive nature of the business questions (specifically the prediction of sales) and aligns with the numeric outcome I want to analyse. It applies the cleansed historical data to detect patterns and relationships, creating predictions and insights (through its coefficients) that can be used by the fictional retail firm’s stock management and demand planning. [3]

References:

[1] M. Singh and Anand Kumar Shukla, “Enhancing Business Intelligence and Decision-Making Through Big Data Analytics,” Nov. 2023, doi: https://doi.org/10.1109/ictacs59847.2023.10389981.

[2] A. Saleem, K. H. Asif, A. Ali, S. M. Awan, and M. A. Alghamdi, “Pre-processing Methods of Data Mining,” IEEE Xplore, Dec. 01, 2014. https://ieeexplore.ieee.org/abstract/document/7027524/ (accessed March. 29, 2025).

[3] A. Jain, P. Gupta, H. K. Saran, D. S. Parmar, Jai Prakash Bhati, and D. Rawat, “Forecasting Future Sales Using Linear Regression Approach,” pp. 269–272, Nov. 2024, doi: https://doi.org/10.1109/cybercom63683.2024.10803255.

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2.1 very good

5.1 correct choice of method